

- 4) Explain the term ‘ontology’ and ‘epistemology’.

1.6 RESEARCH PROCESS

Research process refers to different steps involved in a desired sequence in carrying out research. However, this does not mean that these steps are always in a given sequence. The various steps involved in research can be illustrated in a flow chart as shown in the figure below:

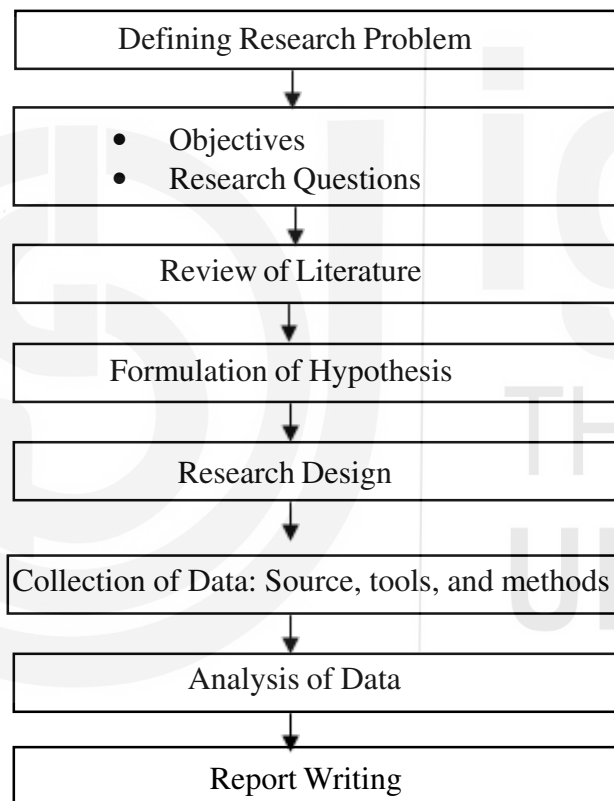


Fig. 1.1: Flowchart of Research Process

The above activities or steps overlap continuously rather than following the prescribed sequence strictly. The steps are not mutually exclusive. The order illustrated is meant only to provide a procedural guideline for research. The steps are briefly elaborated below:

- a) **Defining the research problem:** Selecting and properly defining the research problem is the first foremost step. The problem to be investigated must be defined categorically. It is important to identify the general area of interest or a particular aspect of a subject matter desired to be studied. Initially, the problem may be stated in a broad way and later it can be narrowed down in operational terms. Essentially two steps are involved in formulating the

research problem: (i) Understanding the problem thoroughly; and (ii) rephrasing it into meaningful terms from operational/analytical point of view.

It is better to select the subject that is familiar with easy access to research material and data sources. Apart from the topic, following points need to be stated clearly in the research problem:

- 1) rationale behind the research problem;
- 2) the aims and objectives as per the requirements of the research questions. The statement of the objectives determines the data to be collected, hypothesis to be tested, techniques for data collection and analysis to be adopted, and the relations intended to be explored;
- 3) the research questions in the light of the objectives and the theoretical arguments/foundation on which it rests;
- 4) developing the ideas through discussions; and
- 5) re-phrasing the research problems identified in (i) above into a working proposition.

The **different steps** to be followed while defining the research problem, therefore are:

- statement of the problem first in a general way to be later sharpened with the help of literature review,
- understanding the nature of the problem, and
- surveying the available literature.

In addition to the above, the following points should also be observed while defining the research problem:

- Technical terms and words or phrases used in the research problem should be explicitly defined.
 - Basic assumptions or postulates relating to the research problem need to be clearly stated.
 - A clear and unambiguous statement of the investigation should be provided.
 - The time-period required and the scope of the study must be duly stated.
 - The sources of data and its limitations must be explicitly mentioned.
- b) **Review of Literature:** The review of literature is meant to gain insight on the topic and gain knowledge on the availability of data and other materials on the theme of proposed area of research. The literature reviewed may be classified into two types viz. (i) literature relating to the concepts and theory

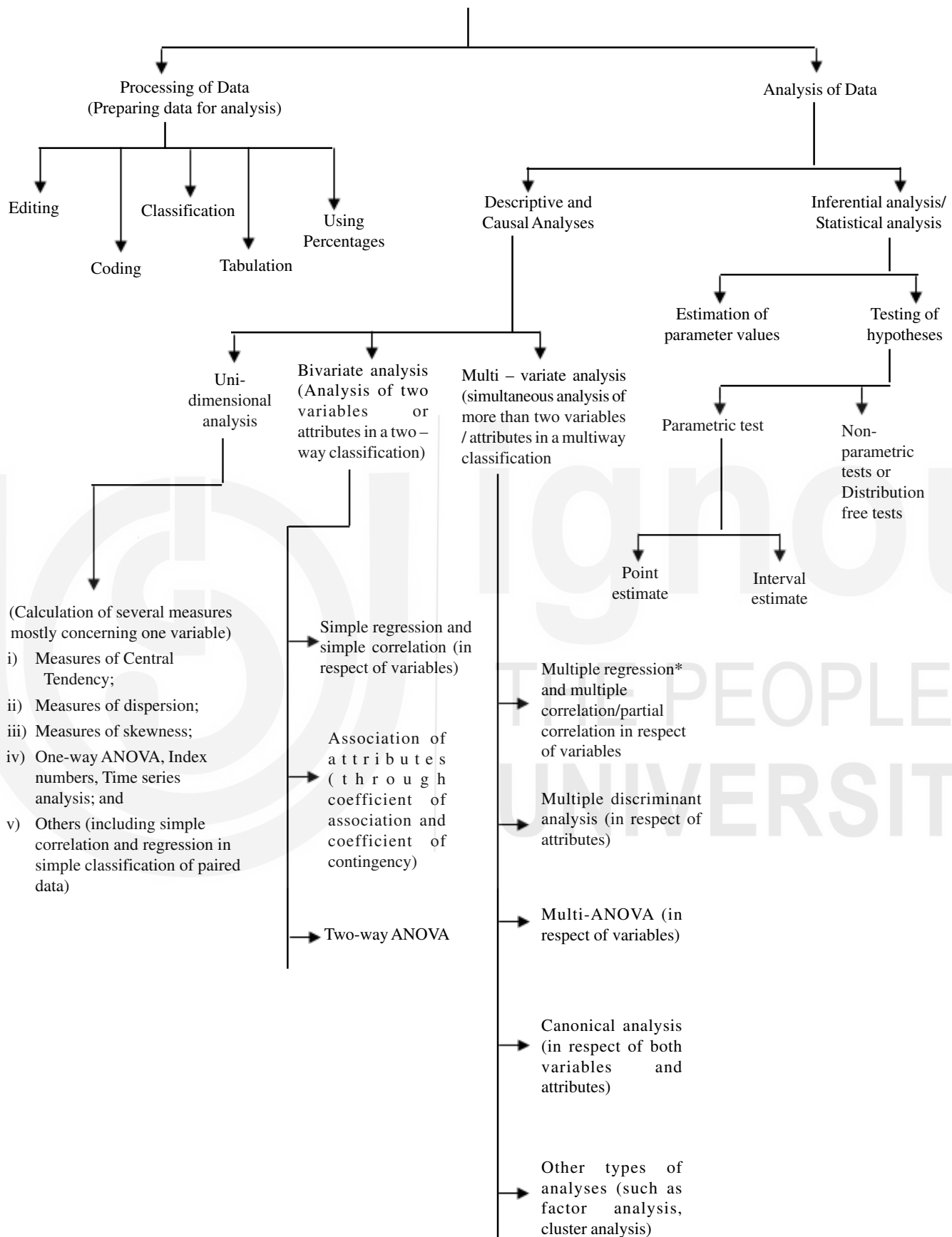
and (ii) empirical literature consisting of findings in quantitative terms by studies conducted in the area. This will help in framing research questions to be investigated. Academic journals, conference proceedings, government reports, books etc. are the main sources of literature. With the spread of IT, one can access a large volume of literature through internet.

- c) **Formulation of Hypothesis:** Specification of working hypothesis (or hypotheses) is the next step of research process. A hypothesis is a tentative statement made which needs to be tested for its logical and empirical confirmation. Hypothesis can be formulated as a proposition or set of propositions providing most probable explanation for occurrence of some event or specified phenomenon. Hypotheses when empirically tested may either be accepted or rejected. A hypothesis must, therefore, be capable of being tested. A hypothesis stated in terms of a relationship between the dependent and independent variables are suitable for econometric treatment. The manner in which hypothesis is formulated is important as it provides the required focus for research. It also helps in identifying the method of analysis to be used.

Prior thinking about the subject, examination of the available data and material related to the study, discussion with colleagues and experts help the researcher in formulation of hypothesis. Exploratory or descriptive research can be carried out even without hypothesis.

- d) **Research Design:** Research design is the logical conceptual structure within which research is conducted. It is the blueprint for the collection, measurement and analysis of data. Detailed discussion on Research Design has been provided in Unit 6 of Block 2.
- e) **Collection of Data:** Collection of data is an essential part of the research process. Data can be primary or secondary. Data collected by the researcher, say by a survey, is primary. The data already collected by some agency and available in some published form is secondary. There are two main techniques of data collection – (i) census and survey (ii) observation method. Primary data can also be collected by means of experiments (e.g., yield under certain conditions, observation at many time points of a certain phenomena, etc.). Intensive fieldwork methods include observation, interview, case study, etc. A survey is usually conducted by the canvassing of a questionnaire. Issues of data collection and sampling design have been discussed in detail in Unit 7 of Block 2.
- f) **Analysis of Data:** Analysis involves steps like categorization, coding, tabulation, etc. The principle for classification or categorization of data has to be based on the problem under study or the hypothesis formulated. The category must be exhaustive and sufficient for classifying all responses. They must be distinct, separate and mutually exclusive. **Coding** involves grouping of responses falling under a particular category. **Tabulation** is a means of organizing the responses to facilitate comparisons bringing up the inherent relations between two or more variables. It is an orderly arrangement of data in columns and rows. Analysis and inference is usually aided by the application of different statistical and econometric techniques. Some of the major techniques usually employed in research have been reflected in the **summary chart 1.2**

**Summary Chart 1.2
ANALYSIS OF DATA**



Most of these analytical techniques have been covered in Units 9,10,11 and 12 of Block 3 and Units 13,14,15,16 and 17 of Block 4 of this course.

Report Writing: Originality and clarity are the two vital components of research report. It is the ultimate test of one's analytical ability and communication skills. It is an exercise involving the organization of ideas. The research report needs to be presented in such a manner that the readers can grasp the context, methodology and findings easily. The report comprise of two parts: the preliminary pages and the main text. In the preliminary pages, the report should indicate the title of the research study, name of the researcher (and his team members) and the name of the institution and/or the month/year of preparation of the report. This should be followed by a 'preface' in which the main context of preparing the report along with key findings must be presented. Towards the end of the 'preface', the important sources/persons can suitably be acknowledged.

The main text begins with an introductory chapter followed by the major aspects of the study organized into different chapters. The introductory chapter should contain a clear statement of the objectives of the study, rationale behind the study, a brief summary of the literature review, hypotheses tested (if any) and the definitions of the major concepts employed in the study. The methodology adopted in conducting the study must also be fully explained along with an explicit mention of the limitations of the study. The subsequent parts of the main text, should present the major aspects of the study arranged in a logical sequence splitted into appropriate sections and subsections. The inter-connection between different sections should be properly maintained so that the report reads smoothly.

The implications of the results of the study should be stated towards the end of the report. The implications may comprise of:

- i) the inferences drawn from the study;
- ii) the conditions which may limit the extent of generalizations of the inferences drawn; and
- iii) the questions that remain unanswered along with new areas for further research identified. The conclusion drawn from the study should be clearly related to the objectives/hypotheses stated in the introductory section.

The report may also include an 'executive summary' outlining the context and methodology, and major findings of the study. The 'executive summary' is placed right at the beginning (i.e. before the introductory chapter) so as to provide a concise picture of the entire report.

1.7 HYPOTHESIS: ITS TYPES AND SOURCES

Hypothesis are potential explanations that can account for our observations of the external world. They usually describe cause and effect relationships between a proposed mechanism or process (the cause) and our observations (the effect).

In quantitative research, there are two methods of hypothesis generation: deductive method and inductive method.

If hypothesis is generated from a theory, it is called deductive approach. On the other hand, if it is generated from observation, it is termed as inductive approach.

Deductive approach: Theory → Hypothesis → Observation → Confirmation.

Inductive approach: Observation → Pattern → Tentative Hypothesis → Theory.

In qualitative research, a hypothesis might be framed in terms of social setting having certain features, which through observation, can be confirmed or falsified. In survey or experimental research or otherwise hypothesis testing establishes the statistical significance of a finding.

1.8 THE NATURE, SOURCES AND TYPES OF DATA

For undertaking any meaningful research in terms of situational assessment, testing of models, development of theory, evaluation of economic policy, data is essential. The availability of data therefore, determines the scope of analysis. In any research, the researcher is expected to state the sources of the data used in the analysis, their definitions, and methods of collection.

The data may be of three types; Time series, Cross-section and Pooled.

- 1) **Time Series Data:** It is a set of observations on the values that a variable takes at different times. Such data may be collected at regular time intervals such as daily (i.e. prices, weather reports etc.), weekly (like money supply figures), monthly (i.e. consumer price index etc.) quarterly (i.e. GDP), annually (i.e. government budget etc.).
- 2) **Cross Section Data:** Cross-section data are data on one or more variables collected at the same point of time. For example the data on the census of population collected by the Registrar General of India.
- 3) **Pooled Data:** In pooled data, the elements of both time series and cross section are clubbed. For example, over a period of time say from 2000 to 2013, we have data on saving, investment and GDP across Indian states.

Panel, Longitudinal or Micro Panel Data: This is a special type of pooled data in which the same cross-sectional (say a family or firm) is surveyed overtime.

The Sources of Data: The data used in empirical analysis may be collected by a governmental agency (e.g. CSO, NSSO, RBI, Labour Bureau etc.), an international agency (e.g. International Monetary Fund (IMF) or a private organization. Such data is called secondary data because these are collected from secondary sources. The details about the kind of secondary data compiled by the different data collecting agencies have been presented in Block 6 of this course.

The data collected by the investigator or researcher through field work is termed primary data. Such data is collected by using different tools like questionnaire, schedule, interview etc. under quantitative approach and participant observation, open ended interview, group discussion, key information etc. (under qualitative approach). The methods for collecting the primary data have been discussed in Unit 7 of Block 2.

1.9 MEASUREMENT SCALES OF VARIABLES

Measurement is the process of observing and recording the observation that are collected. A variable can be measured at four levels: ratio scale, interval scale, ordinal scale and nominal scale. The suitability of analytic technique depends on the measurement scale. A particular econometric/statistical technique that may be suitable for ratio scale variables may not be suitable for nominal scale variables.

It is therefore desirable to know the distinctions among the four types of measurement scales. These are discussed below:

- 1) **Nominal Scale:** Under the nominal scale, the data is recorded into categories, without any order or structure. In other words, if against any question/ statement, response is recorded simply yes/no, then the scale will be nominal. It has no order and there is no distance between yes and no.

The statistical techniques that can be used with nominal scales data are: Mode, cross tabulation – with chi-square. Other highly sophisticated modeling techniques available for nominal scale data are – logistic Linear Regression Model, Principal component analysis, factor analysis etc.

- 2) **Ordinal Scale:** In terms of power of measurement, ordinal scale comes next to nominal scale. Recording Ranks against various options/choices is the simplest ordinal scale. If you are asked by a researcher to rank 5 fruits from most favorable to least favorable, he is essentially asking you to create an ordinal scale of preference.

Median, Mode, rank order correlation, non-parametric correlation and modeling techniques are used with ordinal data.

- 3) **Interval Scale:** In a situation when we not only talk about **differences in order** but also **differences in the degree of order**, it is referred to as interval scale. For example, if we are asked to rate our satisfaction with a piece of software on a 7 point scale, from dissatisfied to satisfied, we are using an interval scale.

Mean and standard deviation, correlation, regression, analysis of variance, factor analysis techniques can be used with interval scale data.

- 4) **Ratio Scale:** A ratio scale is the top level of measurement and satisfies the following properties:
 - i) Measurement of each observation of a variable in numerals (quantitative terms) and hence possible to work out the ratio of two observations. For a variable X taking two values X_1 and X_2 the ratio will be X_1 / X_2 .
 - ii) Measurement of distance between two observation X_1 and X_2 i.e. $(X_2 - X_1)$.
 - ii) Indication of the natural ordering (ascending or descending) of the elements of a variable. Therefore, comparison such as $X_2 \geq X_1$ or $X_2 \leq X_1$ are meaningful.

The statistical techniques used in interval scale can easily be used in ratio scale also. You will find elaborate discussion on measurement scales and scaling techniques in Unit 8 of Block 2.

Check Your Progress 2

- 1) In what way review of literature helps a researcher?

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2) What is meant by a hypothesis?

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3) Distinguish between time series data and cross section data.

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4) State the various measurement scales of data.

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1.10 LET US SUM UP

Research plays a significant role in human progress. It inculcates scientific temper and logical thinking. Research refers to a scientific and systematic enquiry aiming either to discover new facts or to verify the old facts. The theoretical perspectives, tools and techniques of data collection and methods of data analysis together constitute the Research Methodology. Broadly there are two basic approaches of research; quantitative and qualitative. Quantitative approach generates the data in quantitative terms. This can, therefore, be subjected to quantitative analysis. Subjective assessment of attitudes, opinions and behaviour are tackled by the qualitative approach to research. Research process involves seven steps – identification of research problem, review of literature, objectives, formulating the hypothesis, finalizing research design, collection of data, analyzing the data and report writing. Data can be of three types: time series, cross section and panel data. A variable can be measured at four levels: ratio scale, interval scale, ordinal scale and nominal scale. The suitability of analytic technique depends on the measurement scale. Hypothesis is a tentative explanation of any event or phenomenon in the external world. There are two methods of hypothesis generation: deductive method and inductive method. Thus, this unit provides an overview of the conceptual foundation of Research Methodology and enables the learner to comprehend the various processes involved in carrying out the research study.

1.11 KEY WORDS

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| Coding | : A system of symbols, letter or words used in transmitting messages. |
| Epistemology | : Epistemology refers to the theory of knowledge of how human beings come to have knowledge of the world around them – of how we know what we know. Broadly there are two theories: Rationalism and Empiricism. |

- Rationalism** : Rationalism is based on the idea that reliable knowledge is derived from the use of “pure” reason.
- Empiricism** : Empiricism envisages that the knowledge of the world can be obtained only through direct sense-experience.
- Experimental Testing Research** : Research in which independent variables are manipulated.
- Ontology** : It is a branch of philosophy that is concerned with the nature of reality. It deals with the theories about what makes up reality.
- Induction** : Induction is a process for moving from particular statements to general statements. This logic is used in social sciences to produce theory from data.
- Deduction** : Deduction is a process used to derive particular statements from general statements. A hypothesis is deduced from a theory and is tested by empirical data.
- Abduction** : Abduction refers to the process of moving from the way social actors describe their way of life to technical, social scientific description of that social life. It has two stages: (a) describing these activities and meaning and (b) deriving categories and concepts that can form the basis of an understanding or an explanation of the problem at hand.
- Retroduction** : Retroduction refers to the process of going back from, below, or behind observed patterns or regularities to discover what produces them. This logic of enquiry focuses to locate the structures and mechanisms that have produced the regularity. These structures and mechanisms envisage the tendencies or powers of things to act in a particular way.
- Research Hypothesis** : Research hypothesis is a predictive statement that relates an independent variable to a dependent variable. Such a hypothesised relationship is usually meant to be tested by research method. Predictive statements, which cannot be objectively verified, or the relationships that are assumed which cannot be tested do not qualify as research hypothesis.
- Tabulation** : Setting out the data/information in tabular form.

1.12 SOME USEFUL BOOKS

- 1) Kothari C.R. (1985): *Research Methodology – Methods and Techniques*, Wiley Eastern Publication Chapter 1-3 pp 1 to 67.
- 2) LeWis-Beck Michael S; Bryman Aan, Tim Futing, Liao (Ed) (2004): *The Sage Encyclopedia of Social Sciences Research Method*, Volumes 1,2 and 3, Sage Publications, New Delhi.

- 3) David L. Morgan (2014): *Integrating Qualitative and Quantitative Methods*, Sage Publications, New Delhi

1.13 ANSWERS OR HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

- 1) See section 1.2
- 2) See section 1.3
- 3) See section 1.5
- 4) See section 1.3

Check Your Progress 2

- 1) Review of literature enables the researcher to know about the availability of relevant material and data. It also helps in framing the research questions.
- 2) See section 1.7.
- 3) A set of values of a variable at different times is time series data whereas a set of values of one or more variables collected at a point of time is cross section data.
- 4) Various measurement scales of a variable are: ratio scale, interval scale, ordinal scale and nominal scale.